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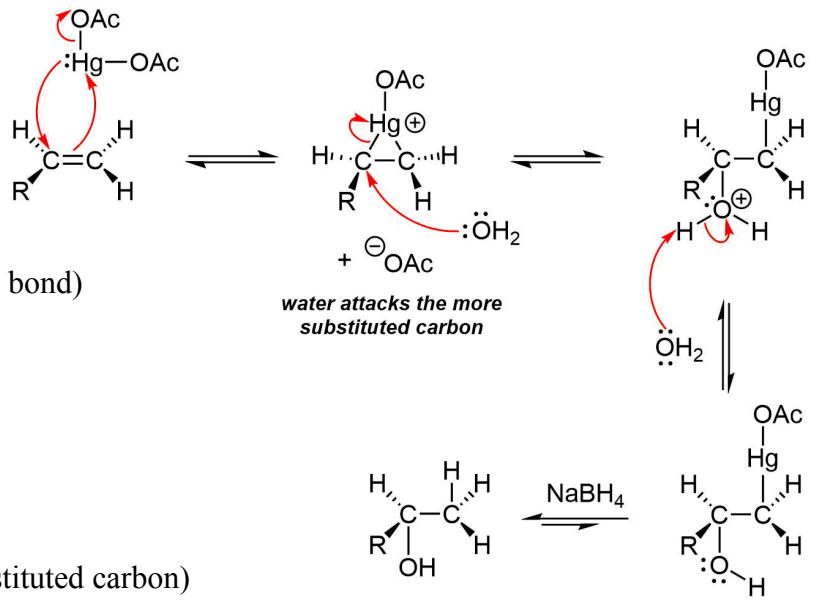
Oxymercuration-Demercuration→Reduction

Brief Description:

The alkene π bond attacks Hg^{2+}, forming a bridged mercurinium ion. Water attacks the more substituted carbon (where positive character is greater), opening the ring. Deprotonation gives an alcohol with Hg attached. $NaBH_4$ then **replaces Hg with H**, completing Markovnikov hydration without forming a carbocation.

Reagent(s):

1. $Hg(OAc)_2$, H_2O
2. $NaBH_4$, NaOH



Product(s)

- Alcohol (OH + H across double bond)

Intermediate(s):

- Mercurinium ion (bridged)

Regiochemistry:

- Markovnikov (OH to more substituted carbon)

Stereochemistry:

- Anti (overall)

Notes:

- No carbocation → no rearrangements

Catalytic Hydrogenation

Brief Description:

The alkene and H_2 both adsorb onto a metal surface. The H–H bond breaks, and both hydrogens are delivered to the same face of the alkene as the π bond is reduced, forming two C–H bonds simultaneously.

Reagent(s):

1. H_2 + Pd/C (or Pt, Ni)

Product(s)

- Alkane (H + H added across DB)

Intermediate(s):

- No intermediate

Regiochemistry:

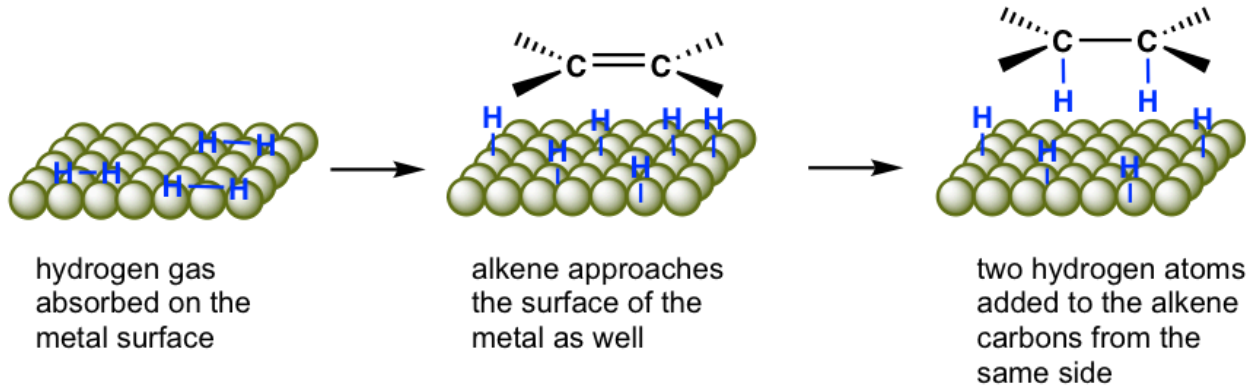
- None

Stereochemistry:

- Syn Addition

Notes:

- Occurs on metal surface; no carbocation. No rearrangements.



simplified diagram for catalytic hydrogenation of alkene

Vicinal Dihalogenation

Brief Description:

The alkene attacks X_2, forming a **halonium ion** and releasing X^-. The halide ion then attacks from the backside, opening the ring and placing the second halogen on the opposite face.

Reagent(s):

1. Br_2 or Cl_2

Product(s)

- Vicinal Dihalide (X & X across DB)

Intermediate(s):

- Halonium ion

Regiochemistry:

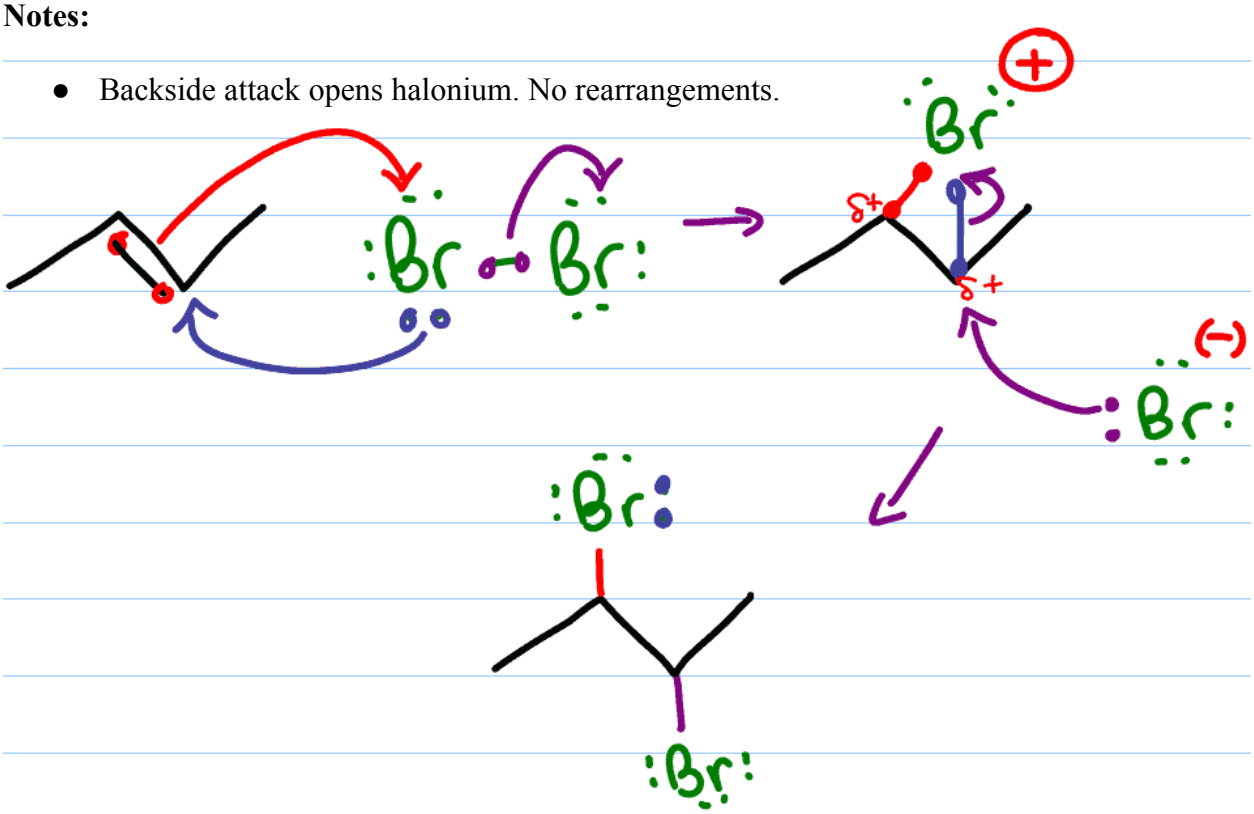
- None

Stereochemistry:

- Anti Addition

Notes:

- Backside attack opens halonium. No rearrangements.



Hydrohalogenation

Brief Description:

The alkene is protonated by HX, forming the **most stable carbocation**. The halide ion then attacks the planar carbocation from either side, giving a racemic alkyl halide.

Reagent(s):

1. HX (HCl, HBr, HI)

Product(s)

- Alkyl Halide (H & X added across DB)

Intermediate(s):

- Carbocation

Regiochemistry:

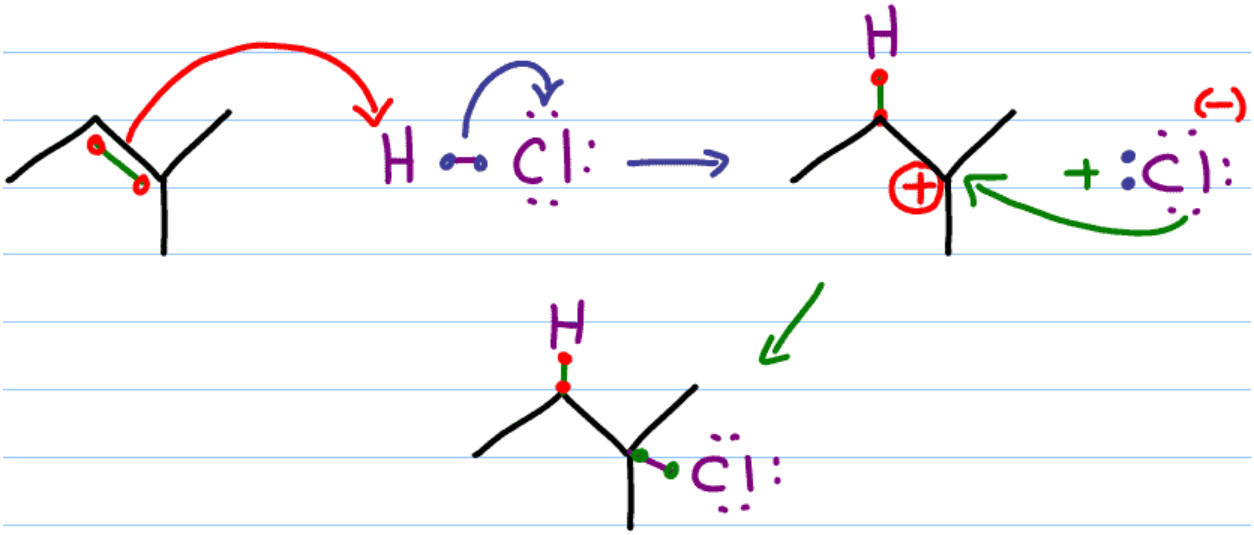
- Markovnikov (X goes to more substituted C)

Stereochemistry:

- Racemic (if chiral center forms)

Notes:

- Rearrangements possible.



Hydroboration - Oxidation

Brief Description:

BH_3 adds across the double bond in a **concerted step**, forming C–B and C–H bonds simultaneously (syn). Boron attaches to the less substituted carbon. In oxidation, H_2O_2/NaOH replaces B with OH without changing the carbon framework.

Reagents:

1. $BH_3 \cdot THF$
2. H_2O_2, NaOH

Products:

- Alcohol (OH & H added across DB)

Intermediate(s):

- Concerted transition state (no intermediate)

Regiochemistry:

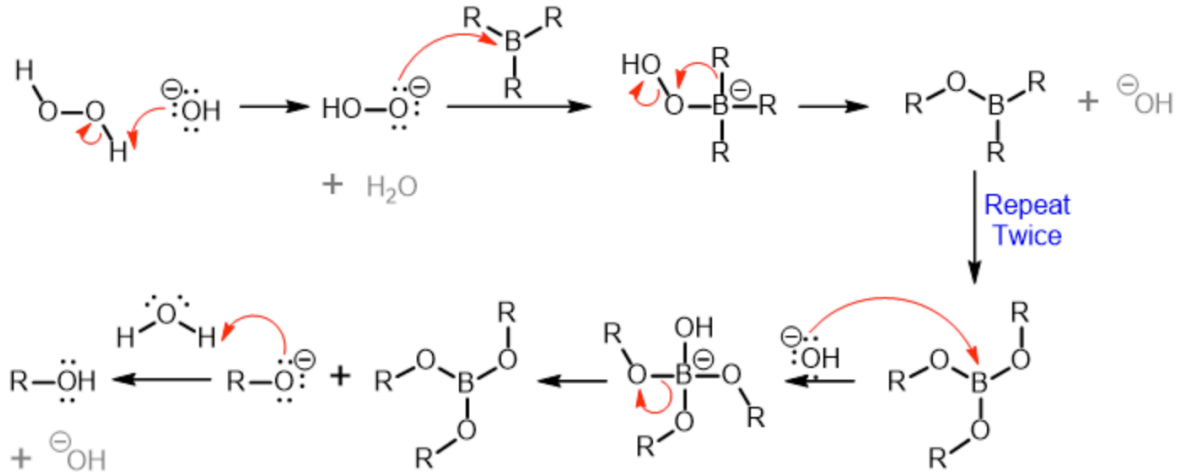
- Anti-Markovnikov (OH to less substituted)

Stereochemistry:

- Syn Addition

Notes:

- No rearrangements. Complicated mechanism–don't need to know.



Acid-Catalyzed Hydration of Alkenes

Brief Description:

The alkene is protonated to form a **carbocation**. Water attacks the carbocation, forming a protonated alcohol, which is then deprotonated to give the alcohol. Rearrangements can occur before nucleophilic attack.

Reagents:

1. $H_2SO_4 + H_2O$ (H_3O^+)

Products:

- Alcohol (OH & H across DB)

Intermediate(s):

- Carbocation

Regiochemistry:

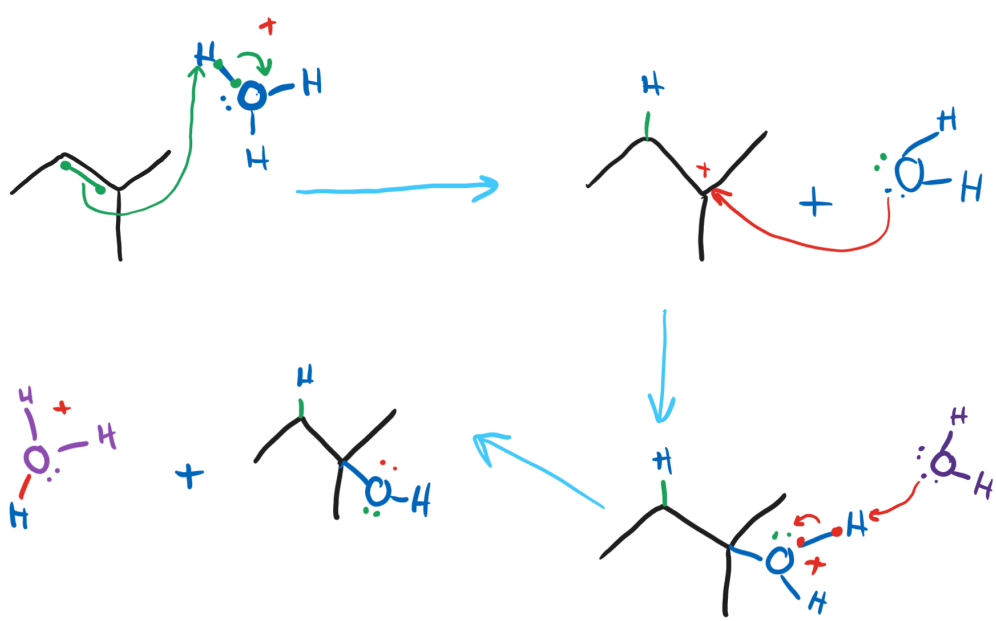
- Markovnikov (OH to more substituted)

Stereochemistry:

- Racemic (if chiral)

Notes:

- Rearrangements possible



Haloalcohol (Halohydrin) Formation from Alkenes

Brief Description:

The alkene reacts with X_2 to form a **halonium ion**. Water (in excess) attacks the more substituted carbon from the backside, opening the ring. Deprotonation gives the halohydrin with OH and X anti.

Reagents:

1. $X_2 + H_2O$

Products:

- Halohydrin (OH & X added across DB)

Intermediate(s):

- Halonium ion

Regiochemistry:

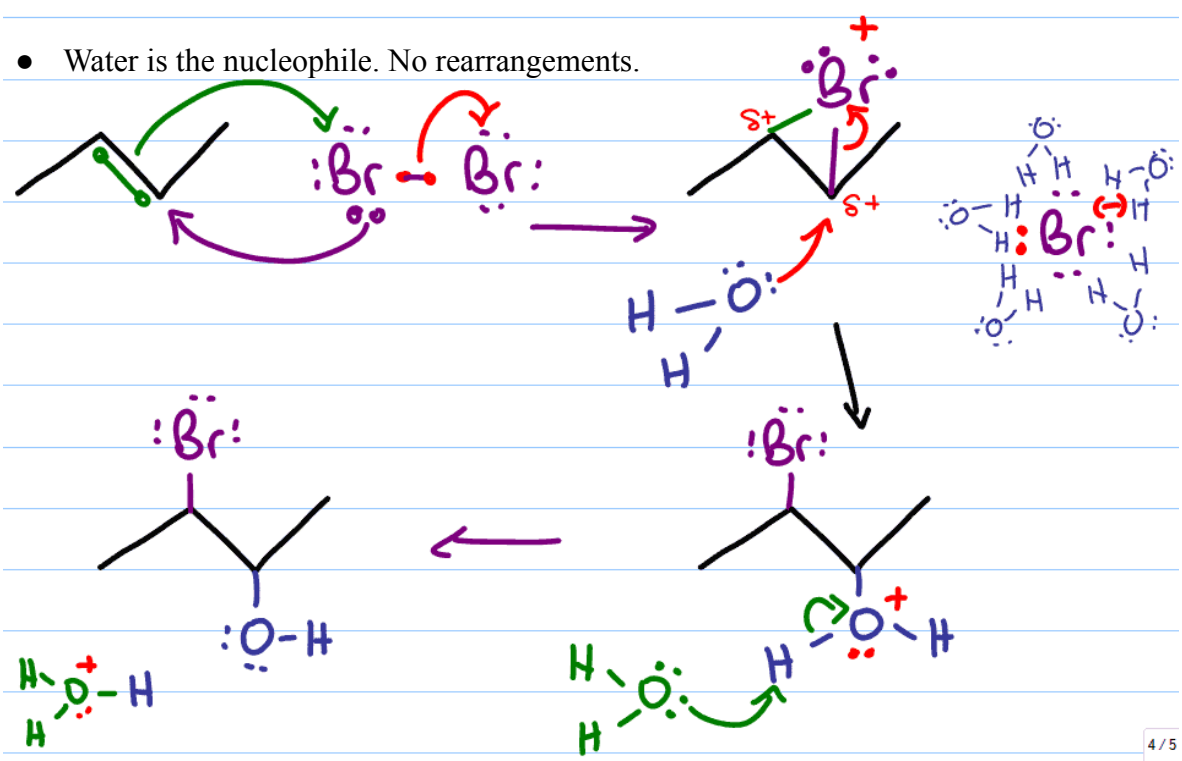
- Markovnikov (OH to more substituted C)

Stereochemistry:

- Anti Addition

Notes:

- Water is the nucleophile. No rearrangements.



Haloether Formation from Alkenes

Brief Description:

The alkene forms a **halonium ion** with X_2. An alcohol (ROH) attacks the more substituted carbon from the backside, opening the ring. Deprotonation gives an ether (–OR) and halogen anti.

Reagents:

2. X_2 + ROH

Products:

- Haloether (OR & X added across DB)

Intermediate(s):

- Halonium ion

Regiochemistry:

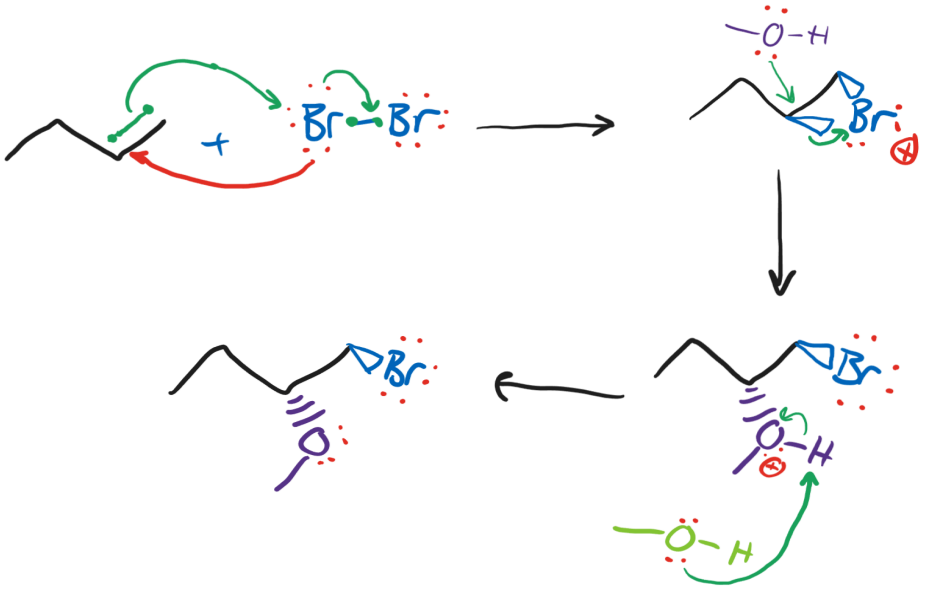
- Markovnikov (OR to more substituted)

Stereochemistry:

- Anti Addition

Notes:

- ROH acts as the nucleophile instead of water. No rearrangements.



Epoxide Synthesis

Brief Description:

A peroxyacid (mCPBA) transfers an oxygen to the alkene in a **concerted step**. The π bond breaks as a new C–O–C ring forms, giving an epoxide with retention of stereochemistry.

Reagents:

1. mCPBA (or peroxyacid) + CH_2Cl_2

Products:

- Epoxide (3-membered cyclic ether) replaces DB

Intermediate(s):

- Concerted (no discrete intermediate)

Regiochemistry:

- None

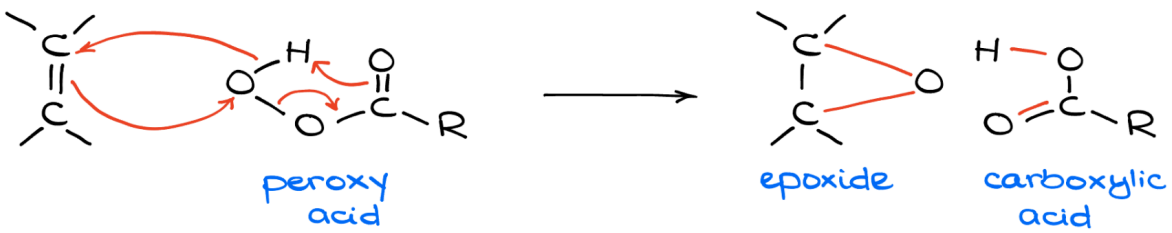
Stereochemistry:

- Syn Addition (retains alkene geometry)

Notes:

- Stereospecific. No rearrangements.

Mechanism:



[Epoxide-Hydrolysis-Mediated] *Vicinal Anti-Hydroxylation* of Alkenes

Brief Description:

The epoxide is protonated, making it more electrophilic. Water attacks the more substituted carbon from the backside, opening the ring. Deprotonation yields a vicinal diol with anti stereochemistry.

Reagents:

1. mCPBA (or peroxyacid), CH_2Cl_2
2. H_3O^+

Products:

- Vicinal diol (OH & OH added across DB)

Intermediate(s):

- Epoxide

Regiochemistry:

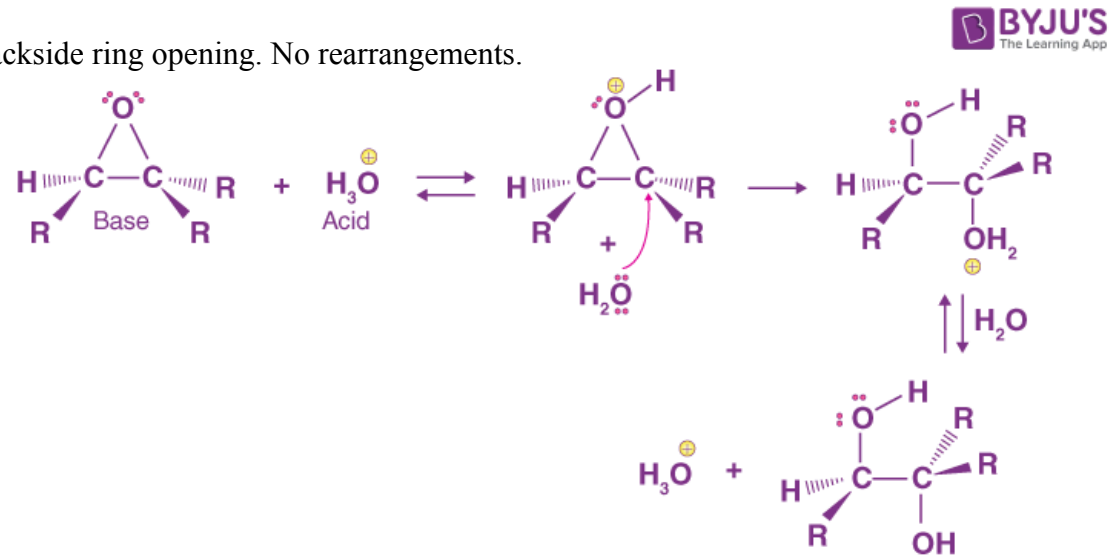
- Water attacks at more substituted carbon (acidic conditions) in an SN2-like fashion, inverting the stereochemistry at the attacked carbon, and opening the epoxide ring.

Stereochemistry:

- Anti Addition

Notes:

- Backside ring opening. No rearrangements.



Vicinal Syn-Dihydroxylation

Brief Description:

The alkene reacts with OsO_4 to form a **cyclic osmate ester**, where both oxygens are added simultaneously to the same face. Reductive workup breaks the Os–O bonds, yielding a syn diol.

Reagents:

3. OsO_4
4. H2S (aq)

Products:

- Vicinal diol (OH & OH added across DB)

Intermediate(s):

- Cyclic osmate ester

Regiochemistry:

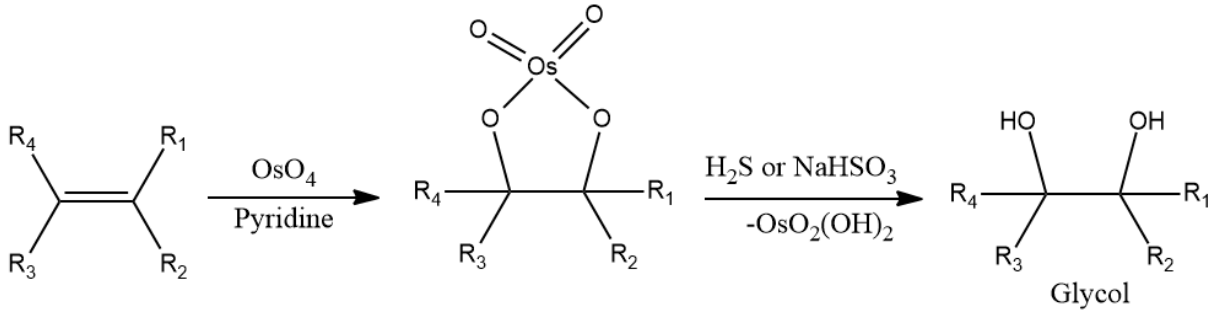
- None

Stereochemistry:

- Syn Addition

Notes:

- Both OH added to the same face. No rearrangements.



Ozonolysis

Brief Description:

Ozone reacts with the alkene to form an unstable **ozonide intermediate**. Reductive workup (Zn + H_2O) breaks the ozonide, splitting the double bond into two carbonyl groups.

Reagents:

5. O_3
6. Zn + H_2O (or $(CH_3)_2S$) [Reductive workup]

Products:

- Aldehydes and/or Ketones

Intermediate(s):

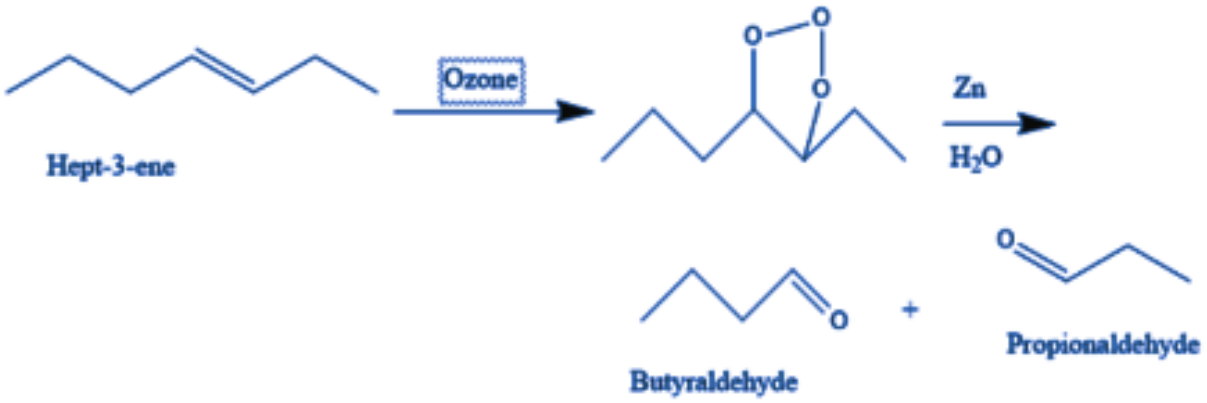
- Ozonide

Regiochemistry:

- Cleavage (N/A)

Stereochemistry:

- N/A

Notes:

- Oxidative workup (H_2O_2 instead of Zn + H2O) → carboxylic acids. No rearrangements.